

Chapter 06: Network Planning & Management

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CS6206 Selected Topics in Wireless Networks



Lecture 6 Learning Objectives

By the end of this lecture, you will be able to:

- Design effective channel plans for 2.4GHz, 5GHz, and 6GHz bands
- Conduct and interpret wireless site surveys
- Calculate AP density based on capacity vs. coverage requirements
- Select appropriate enterprise Wi-Fi architectures
- Configure basic radio resource management (RRM)
- Troubleshoot common wireless interference problems
- Plan for high-density deployments (stadiums, conferences)

The Wireless Network Design Process

- 1 **Requirements Gathering:** What does the network need to do?
- 2 **Capacity Planning:** How many users/devices/applications?
- 3 **Coverage Planning:** Where does coverage need to be?
- 4 **Channel Planning:** Minimizing interference
- 5 **AP Selection & Placement:** Choosing and locating hardware
- 6 **Validation & Optimization:** Testing and adjusting
- 7 **Documentation:** As-built documentation
- 8 **Ongoing Management:** Monitoring and maintenance

Key Insight: Wireless design is iterative - plan, deploy, measure, adjust

Requirements Analysis: Asking the Right Questions

- **User Density:** How many users per area? Now vs. future?
- **Application Requirements:**
 - Bandwidth: Email vs. video conferencing vs. CAD
 - Latency: VoIP (50ms), gaming (100ms), file transfer (no limit)
 - Jitter: Real-time applications sensitive
- **Coverage Areas:** Indoor, outdoor, special areas
- **Mobility Requirements:** Roaming needed? How fast?
- **Security Requirements:** WPA2, WPA3, 802.1X, captive portal
- **Budget Constraints:** Capital vs. operational expenses
- **Regulatory Constraints:** Country-specific rules

Example Requirements: "Support 100 students in a lecture hall, each with 2 devices, streaming 1080p video simultaneously."

Coverage vs. Capacity: The Fundamental Trade-off

Aspect	Coverage-Oriented Design	Capacity-Oriented Design
Goal	Maximize coverage area	Maximize throughput per user
AP Count	Fewer, higher power	More, lower power
Cell Size	Large cells	Small cells
Channel Reuse	Less critical	Critical
Best For	Warehouses, outdoor areas	Offices, classrooms, stadiums
Typical Density	1 AP per 5000+ sq ft	1 AP per 1000-2000 sq ft

Rule of Thumb: Most modern deployments are capacity-limited, not coverage-limited

Because:

1. Everyone has multiple devices
2. Apps consume more bandwidth
3. Dense environments dominate (offices, campuses)

Numerical Example 1: Capacity Planning Calculation I

Problem: Design Wi-Fi for a 100-person conference room (50×80 ft). Each attendee has:

- 1 laptop: 2 Mbps sustained (video conferencing)
- 1 phone: 0.5 Mbps background

Assume 50% protocol efficiency and 70% simultaneous usage factor.

- ① What total throughput is required?
- ② How many 802.11ac APs needed (650 Mbps practical each)?
- ③ What about 802.11ax APs (900 Mbps practical)?

Solution Steps:

- ① Total devices: 100 laptops + 100 phones = 200 devices
- ② Per-device demand: Laptops 2 Mbps, phones 0.5 Mbps
- ③ Total demand: $(100 \times 2) + (100 \times 0.5) = 200 + 50 = 250$ Mbps
- ④ With simultaneity: $250 \times 0.7 = 175$ Mbps

Numerical Example 1: Capacity Planning Calculation II

- ⑤ With protocol efficiency: $175/0.5 = 350$ Mbps required airtime capacity
- ⑥ 802.11ac AP capacity: 650 Mbps practical
- ⑦ APs needed: $350/650 = 0.54 \rightarrow 1$ AP for capacity
- ⑧ But consider: 200 devices on 1 AP = poor performance
- ⑨ Better: 4 APs for device density (50 devices/AP reasonable)
- ⑩ 802.11ax better for high density (OFDMA): 900 Mbps practical
- ⑪ 802.11ax APs: $350/900 = 0.39 \rightarrow 1$ AP capacity, still 4 for device count

Model Answer:

- ① Required throughput: **350 Mbps accounting for overhead**
- ② 802.11ac: **1 AP sufficient for capacity**, but need **4 APs** for device density
- ③ 802.11ax: Same AP count for device density, but **each device gets better experience** due to OFDMA

The 2.4 GHz Band: Limited but Necessary

- **Channels:** 1-13 (varies by country), 1-11 in US
- **Non-overlapping:** Only 1, 6, 11 (or 1, 5, 9, 13 in some regions)
- **Bandwidth:** 20 MHz only (40 MHz not recommended)
- **Range:** Better than 5 GHz (penetrates walls better)
- **Interference:** Severe (microwaves, Bluetooth, Zigbee, etc.)

Design Rules for 2.4 GHz:

- 1 Use only for legacy devices that need it
- 2 Disable 2.4 GHz on some APs in dense deployments
- 3 Use lower power to reduce cell size
- 4 Always use channels 1, 6, 11 (in US) with proper reuse pattern
- 5 Consider disabling 40 MHz channels if enabled

Reality: 2.4 GHz is a "garbage band" - crowded but necessary for compatibility

5 GHz Band: The Workhorse

- **Channels:** 25+ non-overlapping 20 MHz channels
- **(Unlicensed National Information Infrastructure (UNII) bands):**
 - UNII-1 (5150-5250 MHz): Indoor, no DFS
 - UNII-2 (5250-5350 MHz): DFS required (radar)
 - UNII-2 Extended (5470-5725 MHz): DFS required
 - UNII-3 (5725-5850 MHz): Some restrictions
- **DFS (Dynamic Frequency Selection):** AP must detect and avoid radar
- **TPC (Transmit Power Control):** Required in some bands

Design Rules for 5 GHz:

- 1 Use 20 MHz channels for high-density areas
- 2 Use 40/80 MHz only where spectrum is plentiful
- 3 Enable DFS for more channels (but test stability)
- 4 Use band steering to move clients to 5 GHz

6 GHz Band: The Future (Wi-Fi 6E/7)

- **Availability:** 1200 MHz of spectrum (5.925-7.125 GHz)
- **Channels:** 59×20 MHz, 29×40 MHz, 14×80 MHz, 7×160 MHz
- **No Legacy Devices:** Only Wi-Fi 6E/7 devices
- **Three Operation Modes:**
 - **Low Power Indoor (LPI):** Standard indoor APs
 - **Standard Power (SP):** Outdoor/high power, needs AFC
 - **Very Low Power (VLP):** Mobile devices

Features:
Huge spectrum (1200 MHz)
No legacy devices
Very high capacity

Design Implications:

- Massive capacity potential
- Can use wide channels without fear of interference (initially)
- No backward compatibility concerns
- Currently premium pricing

Channel Planning Principles

- 1 **Minimize Co-Channel Interference:** Same channel APs should not hear each other
- 2 **Reuse Pattern:** Like cellular networks
- 3 **Power Control:** Match cell size to coverage area
- 4 **Channel Width Selection:** Wider = faster but fewer channels
- 5 **DFS Consideration:** Will APs switch channels due to radar?

2.4 GHz Pattern (3-channel reuse):

1	6	11	1	6	11
6	11	1	6	11	1
11	1	6	11	1	6

5 GHz Pattern (8-channel example):

36	40	44	48	149	153	157	161
40	44	48	36	153	157	161	149
44	48	36	40	157	161	149	153

Numerical Example 2: Channel Planning for Office Building I

Problem: 3-floor office building, each floor 200×100 ft. Each floor needs 8 APs. Using 5 GHz with 40 MHz channels.

- 1 How many non-overlapping 40 MHz channels available?
- 2 Design channel reuse pattern across floors
- 3 What happens if using 80 MHz channels instead?

Solution Steps:

- 1 40 MHz channels in 5 GHz: Typically 9-12 available depending on DFS
- 2 Assume 10 non-overlapping 40 MHz channels available
- 3 Total APs: $3 \text{ floors} \times 8 \text{ APs/floor} = 24 \text{ APs}$
- 4 Channels needed: 24 APs but only 10 channels ? reuse required
- 5 Reuse factor: $24/10 = 2.4$? need 3-channel reuse pattern
- 6 Design pattern: Assign channels so same-channel APs are maximally separated

Numerical Example 2: Channel Planning for Office Building II

- ⑦ With 80 MHz: Only 4-5 channels available ? reuse factor $24/5 = 4.8$
? need 5-channel reuse
- ⑧ Much more challenging, likely poor performance

Model Answer:

- ① **10 non-overlapping 40 MHz channels** typically available
- ② Need **3-channel reuse pattern** - assign channels so APs on same channel are on different floors and far apart
- ③ With 80 MHz: **Only 4-5 channels** ? severe reuse problem, not recommended for this density

Better Approach: Use 20 MHz channels in high-density office (20+ channels available)

AP Placement Guidelines

- 1 **Avoid Corners:** Center of coverage area is better
- 2 **Consider Ceiling Height:** 8-12 ft ideal, higher requires downtilt
- 3 **Avoid Metal Obstructions:** HVAC, filing cabinets, elevators
- 4 **Separate from Interference Sources:** Microwaves, Bluetooth hubs
- 5 **Consider Aesthetics:** Visible vs. hidden APs
- 6 **Ethical Drop Locations:** Above ceiling tiles, wall mounts
- 7 **Avoid Overlap with Adjacent Buildings:** Coordinate channels

Common Mistakes:

- Placing APs in hallways (creates coverage in hallways, not rooms)
- All APs at same height (creates coverage holes at different heights)
- Ignoring vertical propagation (signals travel through floors)

Site Survey Types

A site survey is the process of planning and designing a wireless network to ensure required coverage, data rates, and roaming capability

- **Predictive Survey:** Software simulation using floor plans
 - Quick, cheap, good for initial planning
 - Less accurate (doesn't account for actual materials)
- **Passive Survey:** Measure existing RF environment
 - Identify interference, existing networks
 - No test APs used A technician walks the site with a receiver but no test Access Points (APs)
- **Active Survey:** Deploy test APs and measure performance
 - Most accurate, most expensive
 - Measures actual throughput, roaming
- **Post-Installation Validation:** Verify deployed network

Tools: Ekahau, AirMagnet, MetaGeek, Wi-Fi analyzers

Active Survey: A technician temporarily installs an AP (often on a tripod) and connects to it. This measures actual throughput (speed) and how well a device "hands off" to another AP while moving (roaming).

RSSI: Received Signal Strength Indicator I

The Foundation of Wireless Connectivity

Definition 0.1

RSSI measures the power level of the signal received by a client device from an access point, measured in dBm (decibel-milliwatts).

Target Values:

- Minimum for data: -65 dBm
- Target for data: -67 dBm or better
- VoIP minimum: -67 dBm
- VoIP target: -62 dBm or better
- High density: -60 dBm or better

Because it's measured in negative numbers, -30 dBm is much stronger than -80 dBm.

Signal Quality:

- -30 to -50 dBm: Excellent
- -50 to -65 dBm: Good

If the RSSI is too low (weaker than -70 or -75 dBm), the device will struggle to stay connected, and data speeds will drop significantly.

RSSI: Received Signal Strength Indicator II

The Foundation of Wireless Connectivity

- -65 to -75 dBm: Fair
- -75 dBm: Poor

Why It Matters:

- RSSI directly determines the maximum achievable data rate
- Weak signals cause retransmissions, increasing latency and reducing throughput
- Client devices may disconnect or refuse to connect below -70 dBm

SNR: Signal-to-Noise Ratio I

Signal Strength Relative to Background Noise

Signal strength isn't everything. You also have to consider the "background noise."

Definition 0.2

SNR is the difference between the received signal strength (RSSI) and the noise floor. It indicates how clearly the signal stands out from background interference.

Calculation

$$\text{SNR (dB)} = \text{RSSI (dBm)} - \text{Noise Floor (dBm)}$$

SNR Targets:

- 40+ dB: Excellent
- 25-40 dB: Good
- 15-25 dB: Fair
- <15 dB: Poor

SNR: Signal-to-Noise Ratio II

Signal Strength Relative to Background Noise

Impact on Performance:

- **25 dB or better:** Required for high data rates
- **15-25 dB:** Lower modulation rates, reduced throughput
- **Below 15 dB:** Unreliable connection, frequent disconnects

Key Insight: A strong RSSI is meaningless if the noise floor is equally high. For example, -50 dBm RSSI with -45 dBm noise gives only 5 dB SNR—a terrible connection despite appearing to have "strong signal."

Noise Floor I

The Silent Killer of Wi-Fi Performance

Definition 0.3

The noise floor is the baseline level of ambient radio frequency energy in an environment, including interference from non-Wi-Fi sources and thermal noise.

Critical Threshold

Noise floor must be below -90 dBm for reliable Wi-Fi operation.

Sources of Noise:

- Bluetooth devices
- Microwave ovens
- Cordless phones
- Radar systems (DFS bands)

This is the "static" in the air caused by microwaves, Bluetooth, or other electronics.

Noise Floor II

The Silent Killer of Wi-Fi Performance

- Industrial equipment
- Adjacent channel interference
- Thermal noise

Noise Level Impact:

RSSI is meaningless without SNR. A signal can be "loud," but if the environment is "noisy" (high interference), the data will be corrupted, leading to retransmissions and lag.

- -95 dBm: Ideal
- -90 to -95 dBm: Good
- -85 to -90 dBm: Concerning
- Above -85 dBm: Problematic

Remediation:

- Identify and eliminate noise sources
- Increase signal strength (move APs closer, add APs)
- Change channels to avoid interference
- Use directional antennas in high-noise environments

Channel Utilization I

How Busy Is Your Airspace?

Definition 0.4

Channel utilization measures the percentage of time a channel is occupied by transmissions, including data, control frames, retransmissions, and interference.

Performance Thresholds

Utilization	Impact
< 30%	Excellent—plenty of capacity
30-50%	Good—normal operating range
50-70%	Warning—performance degradation expected
> 70%	Critical—severe congestion, poor user experience

Channel Utilization

How Busy Is Your Airspace?

What Consumes Channel Time:

- Client data transmissions
- AP beacons and management frames
- Retransmissions (high retry rate)
- Non-Wi-Fi interference
- Overlapping networks (co-channel interference)

How to Reduce Utilization:

- Add more APs (increase capacity)
- Reduce channel width (20 MHz instead of 40/80)
- Move to less congested channels
- Enable band steering to 5/6 GHz
- Limit low-rate clients

Retry Rate I

The Efficiency Metric

Definition 0.5

Retry rate is the percentage of transmitted frames that require retransmission due to not being acknowledged by the receiving device.

Target

Below 10% for good conditions — ideally 5% or lower.

Causes of High Retry Rate:

- Weak signal (low RSSI)
- High interference (low SNR)
- Hidden node problems
- Overloaded AP (too many clients)
- Client driver issues

Retry Rate II

The Efficiency Metric

Impact of Retries:

- Wasted airtime
- Increased latency
- Reduced effective throughput
- Poor user experience (lag, buffering)
- Exponential backoff algorithms make it worse

Key Insight: A 20% retry rate does not mean 20% slower—it often means 50%+ slower because each retry consumes additional airtime, creating a cascade of congestion.

Data Rate & Packet Loss I

The User Experience Metrics

Data Rate:

- The actual throughput achieved by clients
- Should match or approach the expected rate based on signal strength
- Phy rate (advertised) vs. throughput (actual) are different

Typical Data Rate Expectations:

- -65 dBm: 300-800 Mbps (Wi-Fi 5/6)
- -70 dBm: 150-400 Mbps
- -75 dBm: 50-150 Mbps

Data Rate & Packet Loss II

The User Experience Metrics

Packet Loss:

Definition 0.6

Percentage of data packets that never reach their destination.

Critical Threshold

Packet loss must stay below 1% for good performance.

Impact of Packet Loss:

- TCP connections slow dramatically
- VoIP calls become choppy or drop
- Video streaming buffers or fails
- Application timeouts occur

Coverage Targets Summary I

Design Goals by Use Case

Use Case	Minimum RSSI	Target RSSI
General Data	-67 dBm	-65 dBm
VoIP / Real-Time	-67 dBm	-62 dBm
High Density	-60 dBm	-55 dBm
Metric	Minimum	Target
SNR	20 dB	25+ dB
Noise Floor	-85 dBm	-90 dBm
Channel Utilization	70%	<50%
Retry Rate	15%	<10%
Packet Loss	2%	<1%

Coverage Targets Summary II

Design Goals by Use Case

Design Philosophy

These metrics work together: improving one often improves others. For example, increasing RSSI automatically improves SNR and reduces retry rate. A holistic approach to site surveying considers all metrics simultaneously rather than in isolation.

Practical Site Survey Checklist I

What to Validate at Every Location

- ❶ **RSSI:** Measure at edge of coverage area; ensure all locations meet minimum thresholds
- ❷ **SNR:** Check against noise floor; investigate if SNR is low despite good RSSI
- ❸ **Noise Sources:** Scan for interferers; document non-Wi-Fi activity
- ❹ **Channel Utilization:** Monitor during peak usage hours; capture baseline and peak values
- ❺ **Retry Rate:** Test with real clients; high retry often indicates hidden node issues
- ❻ **Roaming Behavior:** Verify clients transition smoothly between APs
- ❼ **Throughput Tests:** Run iPerf or equivalent to validate actual performance

Practical Site Survey Checklist II

What to Validate at Every Location

- ⑧ **Client Mix:** Account for legacy devices that may limit network performance

Remember: "What gets measured gets managed."

Co-channel interference happens when multiple Wi-Fi networks use the SAME channel.

Interference Identification and Mitigation

- **Wi-Fi Interference (Co-Channel):** Other Wi-Fi networks
 - Solution: Channel planning, power control
- **Wi-Fi Interference (Adjacent Channel):** Overlapping channels
 - Solution: Use only non-overlapping channels

Occurs when overlapping channels are used.
- **Non-Wi-Fi Interference:** Microwave, Bluetooth, cordless phones
 - Solution: Spectrum analyzer to identify, avoid those frequencies
- **Self-Interference:** Your own network
 - Solution: Proper channel planning, power control

Spectrum Analyzer View:

- **Wi-Fi:** Bursty, structured patterns
- **Bluetooth:** Frequency hopping, wider bandwidth
- **Microwave:** Continuous wideband noise
- **Zigbee:** Narrowband, constant transmission

Non-Wi-Fi Interference:
Interference from devices not using Wi-Fi protocol but using same frequency band.

Radio Resource Management (RRM)

Radio Resource Management- Set of techniques used in wireless networks to automatically manage and optimize radio resources like channels, transmit power, and coverage.

- **Automatic Channel Selection:** APs scan and choose best channel
- **Automatic Power Control:** APs adjust power to optimize coverage
- **Client Load Balancing:** Distribute clients across APs
- **Client Steering:** Encourage clients to use 5 GHz or roam

RRM Challenges:

- **Oscillation:** APs constantly changing channels/power
- **Client Behavior:** Clients don't always follow suggestions
- **Coordination:** Needs communication between APs

Best Practices:

- Use RRM but with constraints (minimum power, channel blacklists)
- Schedule RRM during off-hours
- Monitor and adjust RRM parameters

Enterprise Architectures Compared

Aspect	Autonomous (Fat AP)	Controller-Based	Cloud-Managed
Management	Individual APs	Central controller	Cloud portal
Scalability	Poor (10s of APs)	Good (1000s)	Excellent
Cost	Low capex, high opex	High capex	Subscription
Features	Basic	Advanced	Advanced
Roaming	Basic L2	Fast roaming	Fast roaming
Redundancy	None	Controller cluster	Cloud redundancy
Example	Home/SMB	Cisco, Aruba	Meraki, Ubiquiti

High-Density Design Principles

- ➊ **More APs, Lower Power:** Small cells reduce clients per AP
- ➋ **Avoid 2.4 GHz:** Disable or use very selectively
- ➌ **Use 20 MHz Channels:** Maximize non-overlapping channels
- ➍ **Client Limits:** 25-50 devices per AP maximum
- ➎ **Airtime Fairness:** Not throughput fairness
- ➏ **Aggressive Roaming:** Encourage clients to best AP
- ➐ **QoS Configuration:** Prioritize real-time traffic
- ➑ **Beacon Interval Adjustment:** Longer intervals reduce overhead

Stadium Example:

- 1 AP per 100 seats
- Directional antennas focused on seating sections
- Disable 2.4 GHz entirely
- Band steering to 5 GHz
- Rate limiting per client

Numerical Example 3: Stadium Wi-Fi Design I

Problem: Design Wi-Fi for a 50,000-seat stadium. Each attendee has 1.5 devices on average. Peak concurrent usage: 40%. Target: 5 Mbps per concurrent user.

- 1 How many concurrent devices at peak?
- 2 What total throughput required?
- 3 How many APs needed (assuming 500 Mbps practical per AP)?
- 4 What about client density (max 40 devices/AP)?

Solution Steps:

- 1 Total devices: $50,000 \times 1.5 = 75,000$ devices
- 2 Concurrent: $75,000 \times 0.4 = 30,000$ concurrent devices
- 3 Throughput required: $30,000 \times 5 \text{ Mbps} = 150,000 \text{ Mbps} = 150 \text{ Gbps}$
- 4 AP capacity: 500 Mbps practical each
- 5 APs for capacity: $150,000/500 = 300$ APs

Numerical Example 3: Stadium Wi-Fi Design II

- ⑥ APs for client density: $30,000/40 = 750$ APs
- ⑦ Use larger number: 750 APs
- ⑧ But $750 \text{ APs} \times 500 \text{ Mbps} = 375 \text{ Gbps capacity} > 150 \text{ Gbps needed}$

Model Answer:

- ① **30,000 concurrent devices** at peak
- ② **150 Gbps total throughput** required
- ③ Need **750 APs** (limited by client density, not capacity)
- ④ This gives **375 Gbps capacity** with headroom for growth

Real World: Stadiums often deploy 1000+ APs for 50,000 seats

Quality of Service (QoS) Planning

- **802.11e/WMM (Wi-Fi Multimedia):** Priority access categories
 - ➊ Voice (AC_VO): Highest priority, shortest AIFS
 - ➋ Video (AC_VI): High priority
 - ➌ Best Effort (AC_BE): Default
 - ➍ Background (AC_BK): Lowest priority
- **Configuration Considerations:**
 - Enable WMM on all networks
 - Map DSCP/IP precedence to WMM categories
 - Set bandwidth limits for certain traffic types
 - Use QoS for upstream (STA→AP) and downstream (AP→STA)

Example VoIP Configuration:

- Voice traffic: AC_VO, maximum priority
- Call Signaling: AC_VI
- Minimum bandwidth guarantee if supported

- **Authentication Methods:**

- WPA2-Personal (PSK): For small deployments
- WPA2/3-Enterprise (802.1X): For organizations
- Captive Portal: For guest access

- **Encryption:**

- CCMP (AES): Mandatory for WPA2
- GCMP: For WPA3 (256-bit)

- **Network Segmentation:**

- Different VLANs for different user types
- Firewall rules between segments

- **Monitoring & IDS/IPS:**

- Rogue AP detection
- Client blacklisting
- Anomaly detection

- **Key Performance Indicators (KPIs):**

- AP/Client counts
- Channel utilization
- Retry rates
- Client RSSI/SNR
- Roaming events

- **Alerting:**

- AP offline
- High utilization
- Security events

- **Regular Tasks:**

- Firmware updates
- Configuration backups
- Performance reviews
- Capacity planning updates

Tools: SNMP, syslog, NetFlow, vendor-specific management

Capacity Planning Formulas

Basic Capacity Formula:

$$\text{APs needed} = \max \left(\frac{\text{Total devices}}{\text{Devices per AP}}, \frac{\text{Total throughput}}{\text{AP throughput}} \right)$$

Airtime Calculation:

$$\text{Airtime per client} = \frac{\text{Data per second}}{\text{PHY rate}} \times \frac{1}{\text{Efficiency}}$$

Channel Reuse Factor:

$$\text{Reuse factor} = \frac{\text{Total APs}}{\text{Non-overlapping channels}}$$

Client Density:

$$\text{Clients per AP} = \frac{\text{Total clients}}{\text{Number of APs}}$$

Note: These are starting points - real world requires testing and adjustment

Common Design Mistakes

- ❶ **Underestimating Device Count:** 2-3 devices per person common
- ❷ **Ignoring Application Requirements:** Video needs more than email
- ❸ **Poor Channel Planning:** Co-channel interference kills performance
- ❹ **No Capacity Planning:** Coverage-only design fails under load
- ❺ **Ignoring Interference:** Non-Wi-Fi interference matters
- ❻ **No Growth Plan:** Networks grow faster than expected
- ❼ **Skipping Site Survey:** Predictive only goes so far
- ❽ **No Monitoring:** Can't fix what you can't measure

- **Band Support:** Ensure APs support 6 GHz (Wi-Fi 6E/7)
- **Multi-Gig Ethernet:** APs need >1 Gbps uplinks
- **PoE Budget:** New APs need more power (PoE+ or PoE++)
- **Cloud Management:** Consider cloud-managed for flexibility
- **AIOps:** AI-driven optimization becoming standard
- **Security:** WPA3 support, enhanced encryption
- **IoT Integration:** Bluetooth, Zigbee radios in APs

Design for: 5-7 year lifespan with technology refresh

- **Capital Expenses:**

- AP hardware
- Switches (PoE capable)
- Cabling
- Controllers (if used)

- **Operational Expenses:**

- Installation labor
- Site survey
- Licensing (controller, support)
- Power consumption
- Maintenance

- **Total Cost of Ownership (TCO):** 3-5 year view

Rule of Thumb: Hardware is 30-40% of total project cost

Case Study: Hospital Wi-Fi Design

- **Requirements:**

- Coverage: All patient rooms, common areas
- Capacity: Medical devices, staff, patients, guests
- Reliability: Critical for medical devices
- Security: HIPAA compliance
- Mobility: Staff roaming with VoIP phones

- **Design:**

- AP in every other patient room (for coverage and capacity)
- Separate SSIDs: Medical devices, staff, guests
- Different VLANs with firewall rules
- WPA2-Enterprise for staff, WPA3 for medical devices
- High availability controllers
- Spectrum management (medical equipment interference)

- **Challenge:** Wireless interference from medical equipment

Summary: Network Planning Checklist

- 1 Requirements documented (users, devices, applications)
- 2 Capacity calculations complete
- 3 Coverage plan with AP locations
- 4 Channel plan for all bands
- 5 Interference assessment complete
- 6 Security plan (authentication, encryption, segmentation)
- 7 QoS configuration planned
- 8 Management/monitoring plan
- 9 Budget and timeline established
- 10 Validation/test plan ready

- **Required Reading:**

- Textbook (Gast): Chapter 6 - "Deploying Wireless Networks"
- Review channel planning examples

- **Optional Reading:**

- CWNP study materials on site surveys
- Ekahau/ AirMagnet white papers

- **Next Lecture (Lecture 7): 802.11a/b/g - The Early Standards**

- Historical context and legacy support
- 802.11b: DSSS, CCK, 11 Mbps
- 802.11a: OFDM, 5GHz band, 54 Mbps
- 802.11g: Bringing OFDM to 2.4GHz
- Compatibility and coexistence issues

Review Questions

- 1 Why is 2.4 GHz channel planning more challenging than 5 GHz?
- 2 Calculate: 500 users, 2 devices each, 30% concurrent, 2 Mbps each ? total throughput?
- 3 When should you use 80 MHz channels vs. 20 MHz channels?
- 4 What's the difference between coverage-oriented and capacity-oriented design?
- 5 Why disable 2.4 GHz in high-density deployments?
- 6 How does DFS affect channel planning?
- 7 What metrics would you monitor to identify interference problems?
- 8 Why might you need more APs for client density than for capacity?

Discussion Question: "You're designing Wi-Fi for a new 40-story skyscraper. Would you use a single ESS across all floors or separate per floor? Consider roaming, broadcast traffic, and failure domains. How would channel planning differ between floors?"

Exercise: Site Survey Simulation

- **Objective:** Plan a wireless network using simulation software
- **Tools:** Ekahau Planner (trial) or open-source alternatives
- **Scenario:** 2-story school building, floor plans provided
- **Tasks:**
 - 1 Import floor plans with wall materials
 - 2 Place APs for coverage and capacity
 - 3 Create channel plan for 2.4 GHz and 5 GHz
 - 4 Generate heat maps for signal strength
 - 5 Adjust plan based on simulation results
- **Deliverable:** Design document with screenshots and justification

Learning Outcome: Experience with predictive site survey tools.

Appendix: Channel Availability by Region

Region	2.4 GHz Channels	5 GHz Non-DFS	5 GHz DFS
USA (FCC)	1-11	8	16+
Europe (ETSI)	1-13	4	19
Japan	1-14	4	8
China	1-13	5	4

Note: Channel availability affects maximum network density possible.

Appendix: AP Density Guidelines by Environment

Environment	APs per 1000 sq ft	Clients per AP	Bandwidth per Client
Office	0.5-1	15-25	5-10 Mbps
Classroom	1-2	20-30	2-5 Mbps
Retail	0.3-0.5	10-20	1-2 Mbps
Warehouse	0.1-0.3	5-15	1-2 Mbps
Healthcare	1-1.5	10-20	2-5 Mbps
Hospitality	0.5-0.8	10-20	2-5 Mbps
Stadium	2-4	40-60	2-5 Mbps

These are guidelines - actual requirements vary based on specific needs.

Appendix: Interference Source Frequencies

- **Microwave Ovens:** 2.45 GHz $\pm$ 50 MHz
- **Bluetooth:** 2.4-2.4835 GHz (FHSS)
- **Zigbee/802.15.4:** 2.4 GHz, 915 MHz, 868 MHz
- **Cordless Phones:** 2.4 GHz, 5.8 GHz, 1.9 GHz (DECT)
- **Baby Monitors:** 2.4 GHz, 900 MHz
- **Video Surveillance:** 2.4 GHz, 5.8 GHz
- **Wireless Peripherals:** 2.4 GHz (mice, keyboards)
- **Radar:** 5.25-5.35 GHz, 5.47-5.725 GHz (triggers DFS)

Mitigation: Spectrum analyzer to identify, then avoid affected channels.

Appendix: Design Validation Test Plan

- ➊ **Coverage Test:** Walk entire area, verify RSSI > -67 dBm
- ➋ **Throughput Test:** Multiple locations, verify target speeds
- ➌ **Roaming Test:** Walk between APs, verify seamless handoff
- ➍ **Capacity Test:** Load network with simulated traffic
- ➎ **Security Test:** Verify authentication, encryption working
- ➏ **QoS Test:** Verify priority traffic gets precedence
- ➐ **Failover Test:** AP/controller failure recovery
- ➑ **Documentation Review:** Verify as-built matches design

Appendix: Budget Template Example

Item	Quantity	Unit Cost	Total
Hardware			
APs	50	\$800	\$40,000
Switches (48-port PoE+)	3	\$2,500	\$7,500
Controller	1	\$10,000	\$10,000
Cabling	50 drops	\$150	\$7,500
Services			
Site Survey	1	\$5,000	\$5,000
Installation	50 APs	\$200	\$10,000
Configuration	1 project	\$8,000	\$8,000
Software/Licensing			
Support (3 years)	50 APs	\$100/yr	\$15,000
Contingency (15%)			\$13,950
Total			\$116,950

Thank you!